培道中學校

姓名：

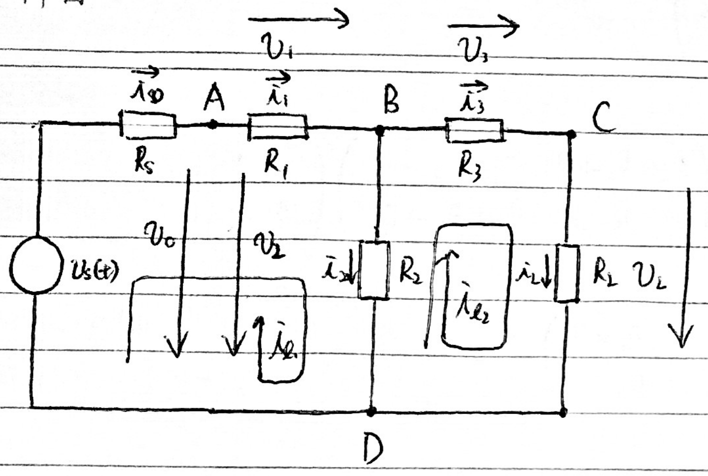
（ 號）

年級：

科目：

成績

1.




支路電壓電流法

$$\left.\begin{aligned} i_0R_s + v_0 &= v_s \\ i_1R_1 - v_1 &= 0 \\ i_2R_2 - v_2 &= 0 \\ i_3R_3 - v_3 &= 0 \\ i_LR_L - v_L &= 0 \end{aligned}\right\}\text{五條支路}$$

$\cancel{i_1R_1 + i_2R_2 - v_0 = 0}$

$$\left.\begin{aligned} i_0 - i_1 &= 0 \\ i_1 - i_2 - i_3 &= 0 \\ i_3 - i_L &= 0 \end{aligned}\right\}\text{結點電流 KCL 方程}$$

$$\left.\begin{aligned} v_1 + v_2 - v_0 &= 0 \\ v_3 + v_L - v_2 &= 0 \end{aligned}\right\}\text{回路電壓 KVL 方程}$$

支路電流法

（同上的三條 KCL 方程）

$$\left.\begin{aligned} i_0R_s + i_1R_1 + i_2R_2 &= v_s \\ i_3R_3 + i_LR_L - i_2R_2 &= 0 \end{aligned}\right\}\text{回路電壓 KVL 方程}$$

回路電流法

$i_{\ell_1} = i_1$ 因 $i_{\ell_1}(R_s + R_1 + R_2) - i_{\ell_2}R_2 = v_s$

$i_{\ell_2} = i_3$ $\quad i_{\ell_2}(R_3 + R_L + R_2) - i_{\ell_1}R_2 = 0$

} KVL 方程

結點電壓法（不妨設 $v_D = 0V$）

$$\left.\begin{aligned} \frac{v_s - v_A}{R_s} &= \frac{v_A - v_B}{R_1} \\ \frac{v_A - v_B}{R_1} &= \frac{v_B - v_C}{R_3} + \frac{v_B - v_D}{R_2} \\ \frac{v_B - v_C}{R_3} &= \frac{v_C - v_D}{R_L} \end{aligned}\right\}\text{KCL 方程}$$

利用回路電流法對方程求解

$$\begin{pmatrix} R_S+R_1+R_2 & -R_2 \\ -R_2 & R_3+R_L+R_2 \end{pmatrix}\begin{pmatrix} i_{\ell_1} \\ i_{\ell_2} \end{pmatrix} = \begin{pmatrix} v_S \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} i_{\ell_1} \\ i_{\ell_2} \end{pmatrix} = \frac{1}{(R_S+R_1+R_2)(R_3+R_L+R_2)-R_2^2}\begin{pmatrix} R_3+R_L+R_2 & R_2 \\ R_2 & R_S+R_1+R_2 \end{pmatrix}\begin{pmatrix} v_S \\ 0 \end{pmatrix}$$

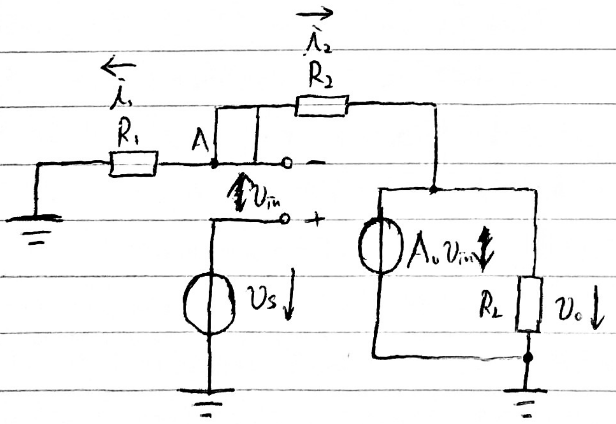
$$= \frac{v_S}{(R_S+R_1+R_2)(R_3+R_L+R_2)-R_2^2}\begin{pmatrix} R_3+R_L+R_2 \\ R_2 \end{pmatrix}$$

$$v_L = i_L R_L = i_{\ell_2} R_L = \frac{R_2 R_L}{(R_S+R_1+R_2)(R_3+R_L+R_2)-R_2^2} v_S$$

$$衰減系数 = \frac{P_{S,max}}{P_{L,rms}} = \frac{\frac{1}{4R_S}\overline{v_S^2}}{\frac{1}{R_L}\overline{v_L^2}} = \frac{R_L}{4R_S}\left(\frac{(R_S+R_1+R_2)(R_3+R_L+R_2)-R_2^2}{R_2 R_L}\right)^2$$

$$= \frac{1}{4}\left(\frac{111^2-35^2}{35\times 56}\right)^2 = 10.05$$

2.




分析A結點

~~$i_1 = i_2$~~ $i_1 + i_2 = 0$

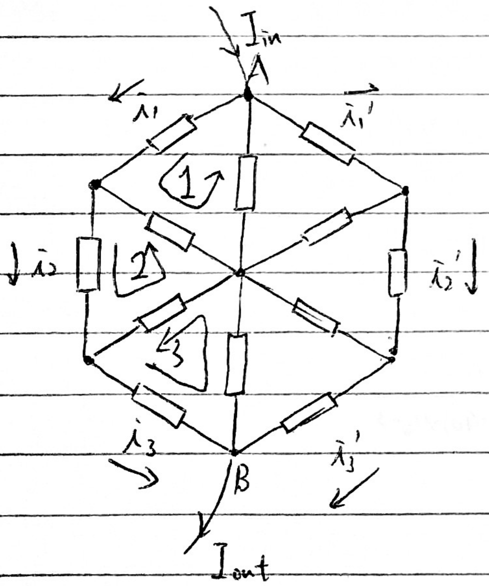
$$\Leftrightarrow \frac{v_S - v_{in}}{R_1} + \frac{v_S - v_{in} - A_0 v_{in}}{R_2} = 0$$

$$\Leftrightarrow v_{in} = \frac{v_S(G_1+G_2)}{G_1+(A_0+1)G_2}$$

$$\Rightarrow v_o = A_0 v_{in} = \frac{A_0(G_1+G_2)}{G_1+(A_0+1)G_2} v_S = \frac{A_0\left(1+\frac{R_1}{R_2}\right)}{1+(A_0+1)\frac{R_1}{R_2}} v_S$$

$\lim\limits_{A\to+\infty} \dfrac{v_o}{v_S} = \dfrac{1+\frac{R_1}{R_2}}{\frac{R_1}{R_2}} = 1+\dfrac{R_2}{R_1}$，即當A充分大時，放大增值只與電阻比有關

3.




不妨假設在A,B端施加電壓 $v = v_A - v_B$ $(>0)$

由對稱性知 $i_1 = i_1'$, $i_2 = i_2'$, $i_3 = i_3'$。又知 $I_{in} = I_{out}$

$$i_1 R + (i_1 - i_2) R + (i_1 + i_1' - I_{in}) R = 0$$

回路1. $4i_1 - i_2 = I_{in}$

$$i_2 R + (i_2 - i_3) R + (i_2 - i_1) R = 0$$

回路2 $i_1 - 3i_2 + i_3 = 0$

$$i_3 R + (i_3 + i_3 - I_{out}) R + (i_3 - i_2) R = 0$$

回路3 $i_2 - 4i_3 = -I_{out} = -I_{in}$

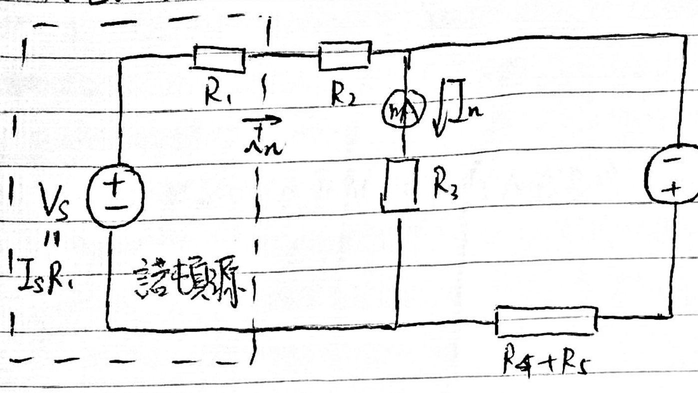
聯立回路1和回路2, $i_1 - 3(4i_1 - I_{in}) + i_3 = 0 \Leftrightarrow 11i_1 - i_3 = 3I_{in}$

聯立回路1和回路3, $4(i_1 - i_3) = 0 \Leftrightarrow i_1 = i_3$

$$\Rightarrow \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 0.3 \\ 0.2 \\ 0.3 \end{pmatrix} I_{in}, \quad I_{in} R_{AB} = v = i_1 R + i_2 R + i_3 R = 0.8 I_{in} R$$

$$\Rightarrow R_{AB} = 0.8R = 0.8\Omega$$

4. 原電路等效化為




其中 $U_n = \begin{cases} 0, & n=1 \\ U_{s_1}=10V, & n=2 \\ U_{s_2}=15V, & n=3 \end{cases}$

$$\begin{cases} V_s = i_n(R_1+R_2)+I_nR_3 \\ U_n = (i_n-I_n)(R_4+R_5)-I_nR_3 \end{cases} \Leftrightarrow \begin{cases} V_s - I_nR_3 = i_n(R_1+R_2) \\ U_n + I_n(R_3+R_4+R_5) = i_n(R_4+R_5) \end{cases}$$

$$\Leftrightarrow (V_s-I_nR_3)(R_4+R_5) = \big(U_n+I_n(R_3+R_4+R_5)\big)(R_1+R_2)$$

$$\Leftrightarrow V_s(R_4+R_5) = \big(U_n+I_n(R_3+R_4+R_5)\big)(R_1+R_2)+I_nR_3(R_4+R_5)$$

$$\Leftrightarrow \frac{V_s(R_4+R_5)}{R_1+R_2} = U_n + I_n\left(R_3+R_4+R_5+\frac{R_3(R_4+R_5)}{R_1+R_2}\right)$$

$\Rightarrow$ 記 $R = R_3+R_4+R_5+\frac{R_3(R_4+R_5)}{R_1+R_2}$, $U = \frac{V_s(R_4+R_5)}{R_1+R_2}$

則 $U_{n+1}-U_n+(I_{n+1}-I_n)R=0$

$$\Leftrightarrow R = \frac{U_n-U_{n+1}}{I_{n+1}-I_n} = \frac{U_1-U_2}{I_2-I_1} = \frac{0-10}{(-60-40)\times10^{-3}} = 100\Omega$$

$$\Rightarrow U = I_1R = 4V$$

因此 $4V = 15V + I_3\times100\Omega \Leftrightarrow I_3 = -110mA$